0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

Draft for discussion · 12 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1 · Fernão de Oliveira
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

Revisions

All alterations to the project scope of work and/or design requirements for the group, against 0.2 Ship Design Project · Baseline B, Azores Feeder: Owner's Requirements (p. 5), are recorded in the following table:

Rev.	Date	Description	By
0	To be set	First issue, after discussion with the students	T. Santos, L. Batista

Baseline B

Deliverable Groups D1–D8 and D10

ID	Deliverable	Status
D1	**Design Basis.** Every requirement tagged Given / Assumed / To be checked against 0.2 Ship Design Project · Baseline B, Azores Feeder: Owner's Requirements (p. 5)	
STU-D1.1	Owner's requirements and port limits: ship type and target capacity, 130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · two loops plus the transit in a week · sequence of port calls (round trip) · relevant characteristics and limits at all eight ports, each with its source	
STU-D1.2	Laws and regulations applicable, incl. environmental targets	
STU-D1.3	Operating speeds: service 11–12 kn, trial, top speed and mainland transit (7 kn)	
D2	**Reference Ship Database.**	
STU-D2.1	Reference ship database with sources: at least 8 small geared container ships and battery ships · other families as your chosen hull needs	
STU-D2.2	Regressions and bounds: analysis of the statistical data	
D3	**Ship Model.** Inputs, outputs, sequence and extent of the calculations (volume carrier) · estimation method selected and named for every item, within its range of validity · outputs that respond correctly to changes in the inputs (qualitative check)	
STU-D3.1	Ship model (computer codes and/or files): hull form coefficients (C_B, C_P, C_M, C_WP, vertical prismatic) · tonnage (GT, NT)	
STU-D3.2	Weight estimation, assumptions and margins: lightship weight (steel, outfit, machinery and superstructure, crane included, with centres) · deadweight components (cargo, batteries as deadweight, lubricating oil, fresh water, provisions and stores, crew) · closed weight balance	

ID	Deliverable	Status
STU-D3.3	Power and propulsion: propulsive coefficients (wake, thrust deduction, relative-rotative and hull efficiency) · propulsive power by Holtrop & Mennen, electric drive, two shafts · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · propeller selection with a cavitation check · side thruster · electric power estimate for each operating condition	
STU-D3.4	Subdivision and capacities: subdivision and compartments (bulkheads, engine room and peaks) · capacities of holds and tanks · containers in the holds and on deck	
D4	**Voyage and Financial Model.** Both loops and the transit · ship model outputs as input	
STU-D4.1	Voyage model: times and energy consumption at sea per leg · times and costs at each port · battery swaps and charging · electricity costs · global results for one round trip and for one year of operation	
STU-D4.2	Financial model: building cost, financing, RFR and NPV	
D5	**Validation Note.**	
STU-D5.1	Validation against reference ships: your model against the reference families you chose, with the choice justified · error per quantity (lightship weight, propulsive and electric generation power, newbuilding cost) · model corrected where needed	
D6	**Optimization Report.** Minimum battery capacity that completes both loops in one week	
STU-D6.1	Optimization report against the design objective, sensitivity analysis: problem definition (design variables, restrictions with their sources, justified objective(s)) · exploration of the domain of the variables · active constraints · optimum against the requirements · delivered code that reproduces the results presented	
STU-D6.2	Code flowchart (ISO 5807): a flowchart of your code, matching it, that shows how the information flows (inputs, each calculation step, the iterations and the outputs), in the symbols of ISO 5807 used correctly	
D7	**Compliance Sheet.** Each applicable rule identified with its source, and each result shown against its limit	
STU-D7.1	EEDI and CII: show why they do not apply, and report energy per TEU-mile	
STU-D7.2	Preliminary intact stability assessment: requirements and typical conditions, incl. ballast conditions · stability while the crane lifts	
STU-D7.3	Draught at every port	
STU-D7.4	Freeboard	
STU-D7.5	Battery system safety (class rules): fire protection, ventilation and gas detection, securing and swapping of the battery containers	
D8	**General Arrangement and Hull Form.**	
STU-D8.1	Hydrostatics check: final main dimensions and hydrostatic properties (displacement, C_B) against the targets, with the error levels discussed	
DWG-D8.1	General arrangement (A3): profile, midship section and main deck, with the main dimensions and the cargo area configuration, the deck container layout, the battery slots aft and the crane reach	
STU-D8.2	Container count on your own arrangement: final cargo capacity (TEU, deadweight) against the targets, with the error level discussed	
STU-D8.3	3D hull form, justified: 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior (files)	
D10	**Datasheet and Presentation.**	
STU-D10.1	Datasheet (template 0.4): one page, A4	
STU-D10.2	Presentation, 15 minutes: oral presentation of the results in PowerPoint, including: objectives of the work (description of the initial concept of the ship); computation model (the computation process, with the adopted methods, criteria and validations); assessment of the results obtained (comparison with existing similar ships); conclusions	
STU-D10.3	Report (0.3 Report Template): a clear description of all the work, a summary of the resulting ship characteristics and a comparison with the statistical data of existing ships of the same market segment	

Specialization scope

One report per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

Deliverable Group D9

ID	Deliverable	Status
D9	**Specialization Documents (2).** One individual report per specialization, with every required study and drawing (below) · methods and assumptions stated · results checked against the acceptance criteria of the specialization	

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

ID	SP01 deliverable	Status
STU-SP01.1	Lightship weight and deadweight distribution along the length	
STU-SP01.2	Four critical loading conditions (subject to justification and/or definition with the instructor)	
STU-SP01.3	Still-water and wave bending moments and shear forces for each loading condition	
STU-SP01.4	Allowable stresses, in bending and in shear	
STU-SP01.5	Preliminary scantlings of the midship cross-section	
STU-SP01.6	Section modulus, moment of inertia and neutral axis of the midship section	
STU-SP01.7	Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them	
DWG-SP01.1	Midship cross-section	

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

The SP02 report is a resistance and propulsion report, justifying:

ID	SP02 deliverable	Status
STU-SP02.1	Detailed ship resistance: each resistance component, the margins and the assumptions, computed from 4 to 14 knots in steps of 1 knot; comparison with the Baseline results. Updated	
STU-SP02.2	Design statement and assumptions; design, machinery and sea margins; efficiencies of the mechanical and electrical equipment	
STU-SP02.3	Optimum Wageningen B-series propeller for the trial condition, the service speed, the top speed and the mainland transit (7 knots)	
STU-SP02.4	Selection of the propeller design condition, with justification; comparison with the Baseline results	
STU-SP02.5	Propeller model description and design features; typical E-motors identified (depending on the type, a gearbox might not be required)	
STU-SP02.6	Generator sizing: one generator able to run both electric motors at the mainland transit speed, with and without the sea margin; typical genset equipment identified	
STU-SP02.7	Motor matching: operating points on the E-motor layout diagram, with the sea margin	
STU-SP02.8	Assumptions on the methodology for the shore–ship electrical interface; battery sizing: calculations, assumptions and high-level procurement specifications	
DWG-SP02.1	Propulsion arrangement: propeller, shaft line (preliminary sizing) and rudder, with the equipment identified	